LMS Desktop App User Manual

This guide walks you through the Library Management System (LMS) desktop application. It's designed for librarians or administrators to manage books, transactions, settings and reports with ease.

1. Overview

The LMS app is built with Electron and provides a clean web-style interface.
It has five main sections accessible via the navigation bar:

- Dashboard – quick stats and recent activity
- Books – add, edit, delete, import, export and scan books
- Transactions – issue, return and renew books
- Reports – view and export printable reports
- Settings – configure library details, backup/restore, templates, manuals

Each page is powered by a set of renderer scripts (`.js` under app folder) that communicate with the backend through `window.electronAPI` (IPC).

2. Getting Started

- Installation
  - - Download the installer from the project release or clone the repository and run `npm install && npm run build` (for developers).
  - - Launch the application; you'll land on the Dashboard (for users).

- Initial Setup
  - - Go to Settings → Library Info.
  - - Fill in library name, school name, borrowing limits, and upload a logo.
  - - Click Save then the page reloads with updated values.

- Database Backup
  - - In the Settings page, use Backup Database to create a time stamped backup file.
  - - Use Reset Database to wipe all records (with confirmation).
  - - Import Books lets you load book records from an Excel template.
  - - Download Template provides the format needed for import; Download Manual saves this systems user guide.

3. Managing Books

- Viewing & Filtering
  - - Open Books page.
  - - Search by title, author, year, category, or book ID.
  - - Filter by category or availability.

- Adding a Book
  - - Click the + Add Book button.
  - - Fill the form (title, author, year, category, copies).
  - - Press Save – the new entry appears immediately.

- Editing/Deleting
  - - Use the Edit or Delete icons on the right side of each row.
  - - Deletion prompts for confirmation.
  - - Edits update the table and maintain active transaction constraints.

- Bulk Operations
  - - Select checkboxes on rows or click the header checkbox to select all.
  - - Click Delete Selected then confirm.

- QR Codes
  - - Click the QR icon to generate a code for the book ID.
  - - File path is shown in a notification.

- Import/Export
  - - Use the header Import button (if available) and the Excel template downloaded earlier.
  - - Export functionality may be present in other modules (not shown).

4. Transactions

- Issuing Books
  - - Go to Transactions page.
  - - Search for member or book.
  - - Select book(s) and click Issue.
  - - Due dates and borrower limits obey settings.

- Returning/Renewing
  - - Find the active transaction.

- - Click Return to mark the book returned.
- - Click Renew to extend the due date (subject to rules).

5. Reports

- Generating
  - - Navigate to Reports.
  - - Choose the report type (daily, monthly, etc.).
  - - Click Generate – a printable table appears.

- Exporting/Printing
  - - Use provided buttons to Print or Save as PDF/CSV depending on implementation.

6. Settings and Utilities

Library Settings – update names, logo, borrowing rules.
Backup & Restore – create backups or upload a `.db` file to restore.
Download Template/Manual – get the Excel template and this guide.
Reset Database – use with extreme caution (erases all record data).

Tips & Troubleshooting

- If the app becomes unresponsive, restart it.
- Always back up the database before major imports or resets.
- Category names are normalized (e.g. `ENGLISH` → `English`) for consistency.
- Notifications appear at the top for success/error messages.
- There's a confirmation modal for destructive actions (delete/reset).

File Structure (for Developers)

main.js – Electron main process entry
preload.js – exposes `electronAPI` to renderer
renderer/ – UI code (HTML, CSS, JS)
assets/js/app/ – page logic
assets/css/ – styles
Utils/ – shared helpers (API, DB, export/import, QR)